WORKED SOLUTION

STEP I 2002, Question 6

Pure • Pyramid Geometry

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Read the question before you go on. It is in the STEP Support Programme's question database:

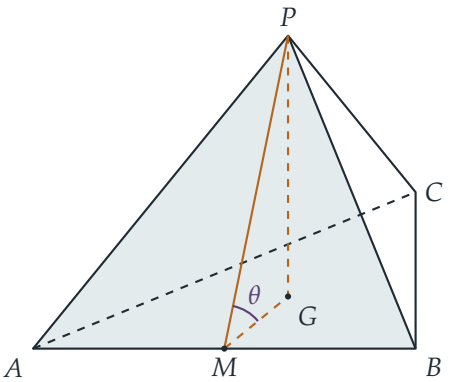
step.maths.org/questions/02-s1-q6

The key idea

By symmetry the apex sits directly above the centre of the equilateral base, so the height comes from one application of Pythagoras. Once we know the volume, we never need to find the new height directly: the volume is the same whichever face is on the ground, so $V = \frac{1}{3} \times (\text{area of the new base}) \times h$. The angle in the final part is the angle between two faces along their common edge, and the cleanest way to see it is in the cross-section through the apex that is perpendicular to that edge.

Solution

Label the base ABC, with sides a, and the apex P, with $PA = PB = PC = b$. Let G be the centre of ABC and M the midpoint of AB.



The volume

Let H be the height of P above the base and F the foot of the perpendicular from P. The right-angled triangles PFA, PFB, PFC share the side $PF = H$ and have hypotenuse

b, so

$$FA = FB = FC = \sqrt{b^2 - H^2}.$$

So F is equidistant from the three vertices: it is the circumcentre of ABC, which for an equilateral triangle is its centre G. The median of ABC has length $\frac{\sqrt{3}}{2}a$, and G lies two-thirds of the way along it from a vertex, so $GA = \frac{2}{3} \cdot \frac{\sqrt{3}}{2}a = \frac{a}{\sqrt{3}}$. By Pythagoras in triangle PGA,

$$H^2 = b^2 - \frac{a^2}{3} = \frac{3b^2 - a^2}{3}.$$

The base has area $\frac{1}{2}a^2 \sin 60^\circ = \frac{\sqrt{3}}{4}a^2$, so

$$V = \frac{1}{3} \cdot \frac{\sqrt{3}}{4}a^2 \cdot \frac{\sqrt{3b^2 - a^2}}{\sqrt{3}} = \frac{1}{12}\, a^2 \left(3b^2 - a^2\right)^{\frac{1}{2}},$$

as required. (A pyramid of this shape exists only when $3b^2 > a^2$, which is exactly when this is real and positive.)

The new height

Now face PAB lies on the ground. It is isosceles with $PA = PB = b$ and $AB = a$, so PM is perpendicular to AB and, by Pythagoras in triangle PMA,

$$PM^2 = b^2 - \frac{a^2}{4} = \frac{4b^2 - a^2}{4}, \qquad \text{area of } PAB = \frac{1}{2}\, a \cdot PM = \frac{a}{4}\sqrt{4b^2 - a^2}.$$

Ask yourself Do we really need to locate the new top vertex to find h?

No. The solid hasn't changed, so its volume is still V, and the volume formula holds for any face taken as the base, with the height measured perpendicular to that face. So

$$V = \frac{1}{3} \cdot \frac{a}{4}\sqrt{4b^2 - a^2} \cdot h.$$

Equating with the first part,

$$\frac{a}{12}\sqrt{4b^2 - a^2}\, h = \frac{a^2}{12}\sqrt{3b^2 - a^2} \quad \Longrightarrow \quad h = \frac{a\sqrt{3b^2 - a^2}}{\sqrt{4b^2 - a^2}},$$

and squaring gives

$$h^2 = \frac{a^2(3b^2 - a^2)}{4b^2 - a^2},$$

as required. Here $4b^2 - a^2 > 0$ because $4b^2 > 3b^2 > a^2$.

The angle of the equilateral face

In the new position the ground is the plane of PAB, and the equilateral face ABC meets it along the edge AB. The angle between two planes is the angle between lines, one

in each plane, drawn perpendicular to their common edge at the same point. At M, the line MP in face PAB is perpendicular to AB (as PAB is isosceles), and the line MC in face ABC is perpendicular to AB (as ABC is equilateral), and G lies on MC. So the required angle is $\theta = \angle PMC = \angle PMG$.

Triangle PMG has a right angle at G, since PG is perpendicular to the whole base. Also GM is a third of the median, $GM = \frac{1}{3} \cdot \frac{\sqrt{3}}{2}a = \frac{a}{2\sqrt{3}}$. Therefore

$$\tan\theta = \frac{PG}{GM} = \frac{\sqrt{3b^2 - a^2}}{\sqrt{3}} \cdot \frac{2\sqrt{3}}{a} = \frac{2\sqrt{3b^2 - a^2}}{a},$$

so the angle between the equilateral triangle and the horizontal is

$$\theta = \tan^{-1}\left(\frac{2\sqrt{3b^2 - a^2}}{a}\right).$$

This angle is acute, because G lies inside the base, so it really is the angle between the planes and not its supplement.

Equivalently, using the new height: the vertex C is at height h above the ground and at distance $MC = \frac{\sqrt{3}}{2}a$ from the edge AB along the face, so $\sin\theta = h / \frac{\sqrt{3}}{2}a$, giving

$$\sin\theta = \frac{2\sqrt{3b^2 - a^2}}{\sqrt{3(4b^2 - a^2)}}, \qquad \cos\theta = \frac{a}{\sqrt{3(4b^2 - a^2)}},$$

which agree with the value of $\tan\theta$ above.

Checks

For a regular tetrahedron, $a = b = 1$: the formulas give $V = \frac{\sqrt{2}}{12}$, $h^2 = \frac{2}{3}$ (the same as the original height, as it must be by symmetry), and $\cos\theta = \frac{1}{3}$, the familiar dihedral angle $\theta \approx 70.53°$ of the regular tetrahedron. As $b \to a/\sqrt{3}$ the pyramid flattens and $\theta \to 0$; as $b \to \infty$ it becomes a tall spike and $\theta \to 90°$. We also built the pyramid in coordinates for 20,000 random pairs (a, b) and computed V from a triple product, h as the distance from C to the plane PAB, and θ from the normals; all three agreed with the formulas to about 10^{-13}. For instance $a = 2$, $b = 3$ gives $h \approx 1.6956$ and $\theta \approx 78.22°$.

Common slips

- Taking GA to be half the median, or using the inradius $\frac{a}{2\sqrt{3}}$ where the circumradius $\frac{a}{\sqrt{3}}$ is needed (and the other way round in the last part).
- Measuring the angle between the face and the ground with a line that isn't perpendicular to the common edge, such as AC or PC.

Beyond STEP

Computing the same volume with different faces as base is a first glimpse of the idea that volume does not depend on how we slice or orient a solid. In a first-year course the volume of a tetrahedron becomes $\frac{1}{6}|\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})|$, a scalar triple product (a 3×3 determinant), and the angle between faces is the angle between their normal vectors, which is how our numerical check worked. Strang's *Introduction to Linear Algebra* develops determinants as volumes.

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